



BASICS OF SIGNAL PROCESSING

CONTENTS

- DISCRETE-TIME SIGNALS AND SYSTEMS
- IIR/FIR SYSTEMS
- Z-TRANSFORM AND STABILITY
- FREQUENCY RESPONSE AND VARIOUS TRANSFORMS
- SAMPLING THEOREM
- FILTER DESIGN



Textbook:

- A.V. Oppenheim and R.W. Schaffer,
Discrete-time signal processing.
Englewood Cliffs, NJ: Prentice-Hall, Inc., 1989.

References:

- S. C. Chan,
ELEC6026 Digital Signal Processing Lecture Notes.
Dept. of Electrical and Electronic Engineering,
The University of Hong Kong, 2010.



IMPORTANT

THREE STAGES OF LEARNING

1

Remember the names!

2

Understand concepts
Adopt from others

3

Write codes
Start from 0



THREE STAGES OF LEARNING

1 Remember the names!



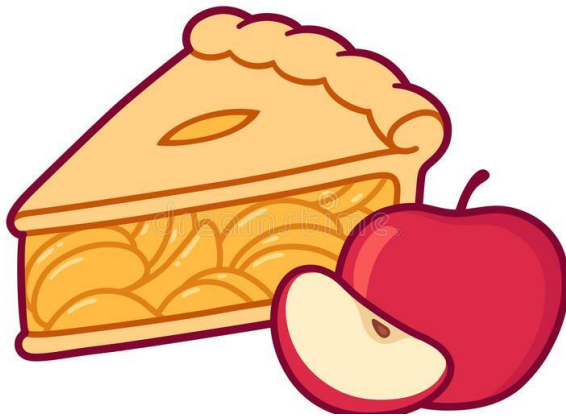
A for Apple

Apple $\in$ Set of Fruits

THREE STAGES OF LEARNING

2 Understand how things work

How to apply
Adopt from others



- E.g. you can bake an apple pie with someone's recipe.



#GordonRamsay #Cooking

Apple and Cranberry Crumble | Gordon Ramsay

Youtube is always your friend!

THREE STAGES OF LEARNING

3 Start from stretch (not easy)



The image is a parody of a news broadcast. It features a desert landscape with a cactus and a butte. In the bottom right corner, Stan Marsh from South Park is standing, wearing his signature blue beanie with a red pom-pom and a red shirt. A large speech bubble from him says "Please grow your apples" in red text. In the top left corner, there is a small inset image of a man with glasses and a purple tie, looking shocked with his hands up, and a speech bubble saying "OMG!". The background is a light blue sky with a green gradient. The overall theme is about growing apples in a desert.

TOP NEWS

NOW PLAYING
Stan learnt A for Apple

Can Stan grow his own apples?

STAN'S CHALLENGES

LIVE STAN IS GROWING APPLES IN THE SAHARA DESERT

Last year I dug a well



Assignments

- **1 Written Assignment**
- **1 Computer based assignment**



Exam (Closed book)

- **Study Problem sets**
- **Programming not examined**

Study the written assignment !



IMPORTANT

When I say it's not examined
Then it's really not examined

Remember this girl !

Only **A** is examined



Not Examined





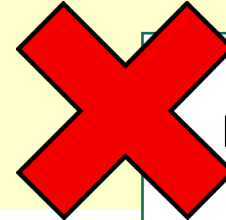
Won't appear in exam

IMPORTANT

APPENDIX A. DERIVATION OF DUALITY (NOT EXAMINED)

- α gives the weights of the training points, aka support vectors.

$$\begin{aligned}\nabla_w L(\mathbf{w}, w_0, \alpha) &= \begin{bmatrix} \frac{\partial}{\partial \mathbf{w}} \left(\frac{1}{2} \|\mathbf{w}\|_2^2 \right) - \frac{\partial}{\partial \mathbf{w}} \sum_{i=1}^N \alpha_i [z_i (\mathbf{w}^T \mathbf{x}_i + w_0) - 1] \\ - \frac{\partial}{\partial w_0} \sum_{i=1}^N \alpha_i [z_i (\mathbf{w}^T \mathbf{x}_i + w_0) - 1] \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{w} - \sum_{i=1}^N \alpha_i z_i \mathbf{x}_i \\ - \sum_{i=1}^N \alpha_i z_i \end{bmatrix} = \mathbf{0}\end{aligned}\tag{A.4}$$



Not Examined



- From (A.4), we have

$$\mathbf{w} = \sum_{i=1}^N \alpha_i z_i \mathbf{x}_i \text{ and } \sum_{i=1}^N \alpha_i z_i = 0.\tag{A.5}$$

- Substitute (A.5) into (A.3), we have

It's optional ! Can skip if not interested

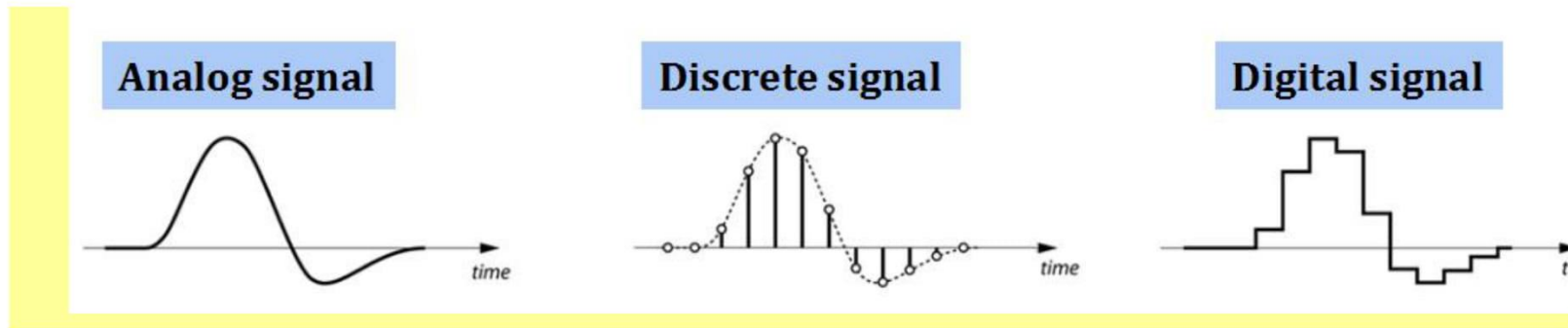


1. INTRODUCTION

Continuous-time signals: signals defined along a continuum of times $x(t)$.

Discrete-time signals: signals defined at discrete times $x[n]$

Digital signals: both time and amplitude are discrete
($x[n]$ amplitude quantized to say 16-bits)



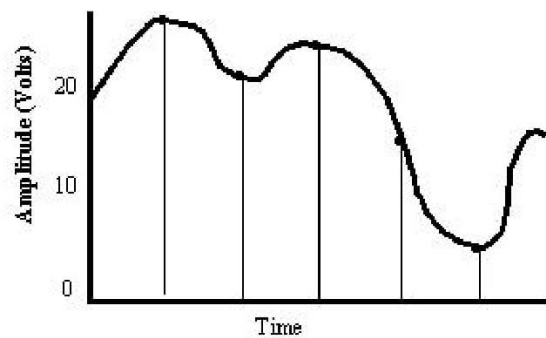


1.1 Discrete-time signals

- Discrete-time signals are obtained by taking samples from continuous-time signal:

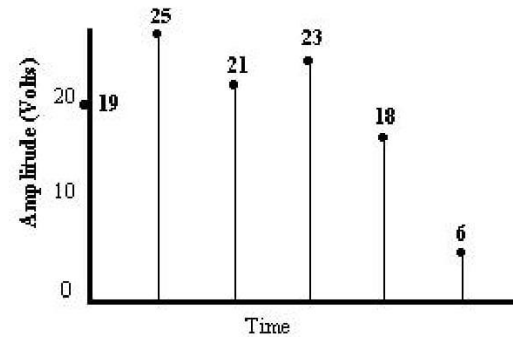
image source <http://www.dilettantesdictionary.org/defimages/analogtodigital.gif>

Fig. 2.1 Continuous-time Signal



$$x_a(t)$$

Fig. 2.2 Discrete-time signal



$$x[n] = x_a(nT), \text{ where } n \text{ is an integer.} \quad (2.1)$$

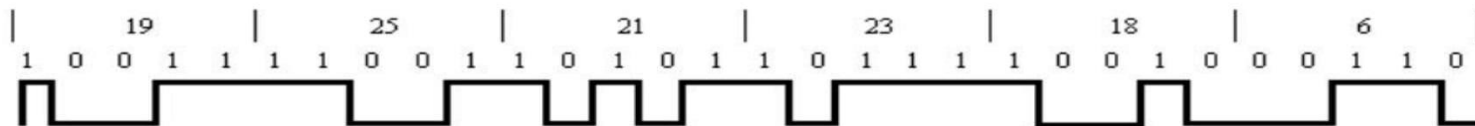


Fig. 2.3 Digital Signal after quantization of Discrete-time signal in Fig. 2.2.



- The discrete-time signal is then round off to the nearest bits.
- This process is called **quantization**.

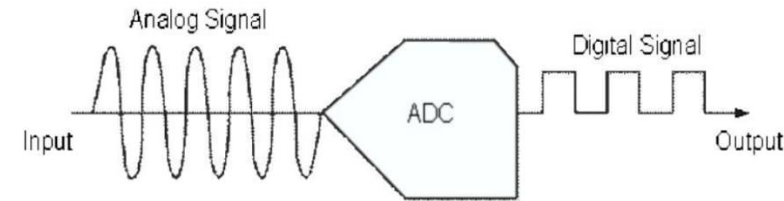
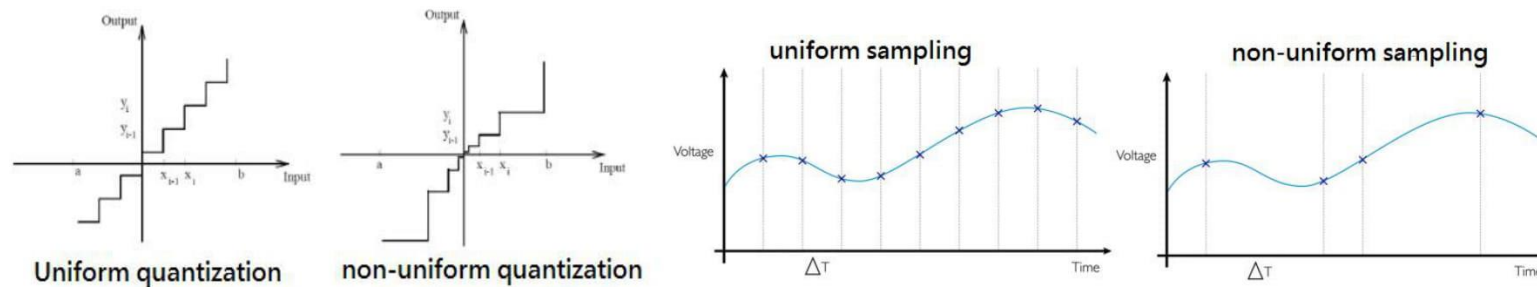


Fig. 2.4 Analog to Digital Converter (ADC)

image source:

https://wiki.analog.com/_media/university/courses/electronics/text/chptr20-f1.gif?w=650&tok=6b5353



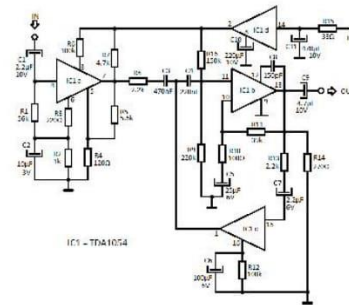
- An ADC converts continuous-time signal to discrete-time signal.
- Both the sampling and quantization processes could be non-uniform.



1.2 Motivation of Digital Signal Processing

Analogue Circuit

Analogue Audio Compressor

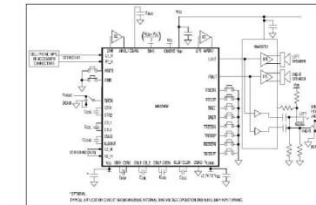
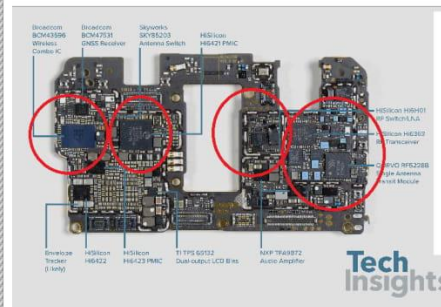


1. Bulky (Large in size)
2. Circuit is **fixed**, difficult to modify its function
3. more liable to design faults, **aging parts** and outside powers such as electrical noise

■ Digital Circuit is quite powerful.

Digital Circuit (DSP Processor)

Huawei Mate 10 circuit



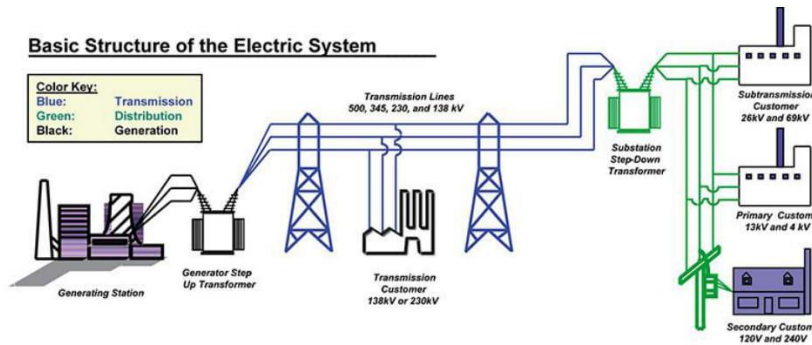
Microprocessors / DSPs !!!

1. Smaller in size
2. A **microprocessor / DSP** can perform multiple functions
3. **Functionality can be modified** by programming.
4. Less vulnerable to faults due to aging



Are Analogue circuits still used ?

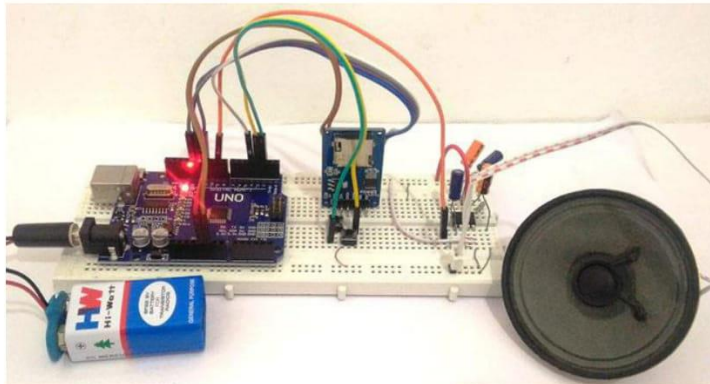
■ Yes !! We still rely on analogue circuits.



■ Power systems

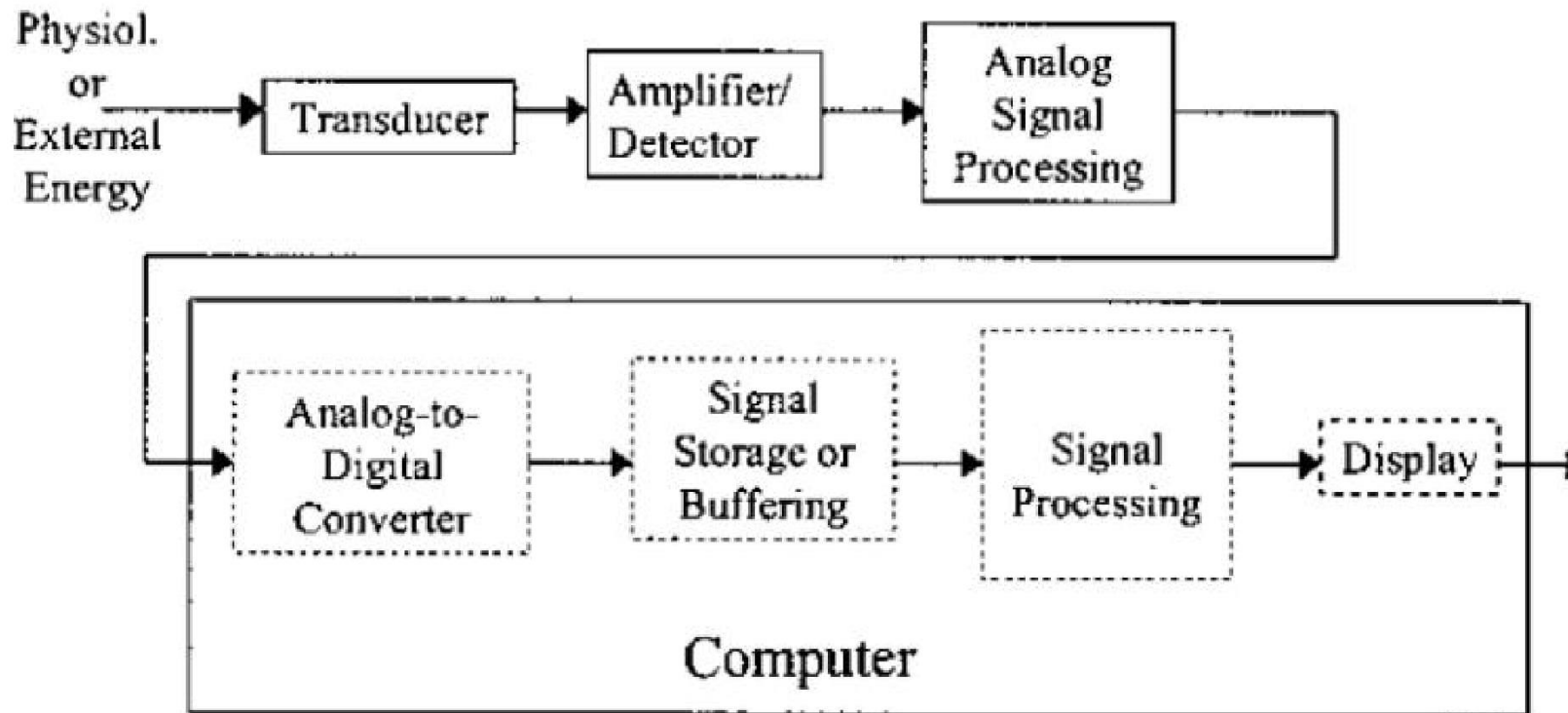
■ Some digital circuits may require an analogue output.

E.g. The arduino can be programmed to playback wav files by driving the speaker using PWM.





Typical Biomedical measurement system



Example 2.1 Analog-to-Digital Converter (ADC) in ECG Holter system Frontend

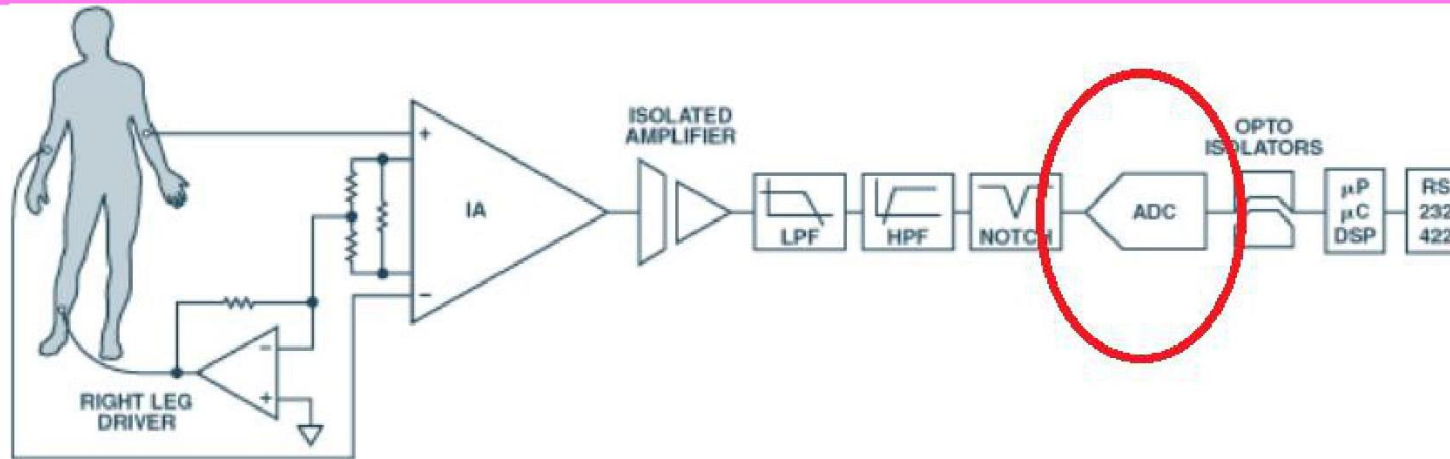


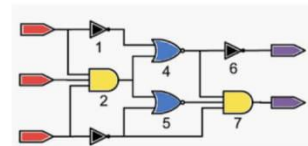
Image source: <https://www.analog.com/-/media/images/analog-dialogue/en/volume-37/number-4/articles/ecg-front-end-design-simplified/ecg-02.gif?la=en>

- **Continuous-time signal (voltage) is picked up between the two electrodes connecting to the non-inverting (+) and inverting (-) inputs of the instrumental amplifier (IA).**
- **The signal is then amplified, filtered before it is passed to an ADC.**



1.3 Analogue system vs Digital System

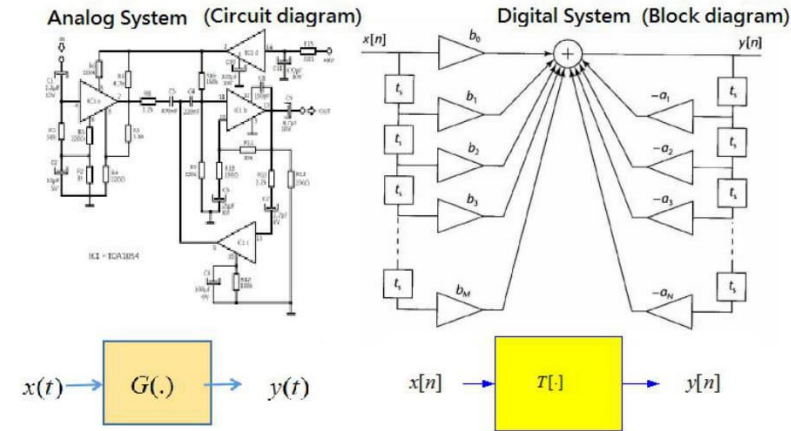
- An analog circuit with input and output signals can be viewed as a **black-box** with input $x(t)$ and $y(t)$.
- Similar applies to digital system.



Digital Circuit



DSP

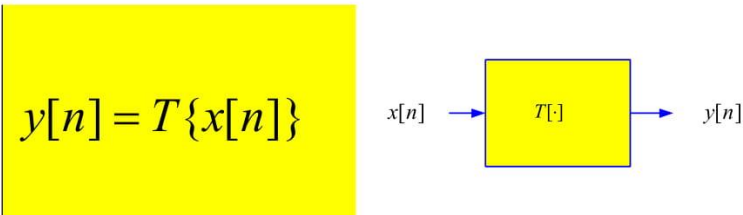


- Digital system are often implemented using a **digital signal processor (DSP)** or even general purpose computer.
- The procedure/operations that occurs in the DSP can be presented as a **block diagram**. (Do not mix up with digital circuit)



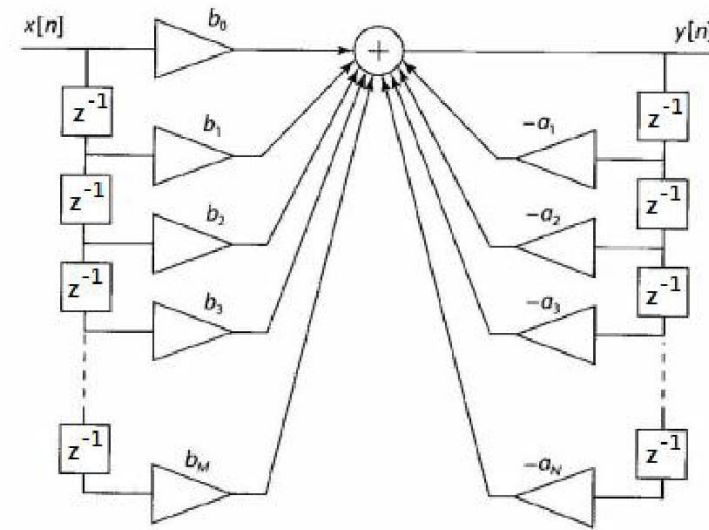
1.4 Discrete-time systems

A discrete-time system is defined mathematically as a **transformation or operator** that maps an input sequence with values $x[n]$ into an output sequence with values $y[n]$.



Major functions of a circuit:

1. **Addition:** Add different signals together
2. **Amplification:** Amplify the signal
3. **Filtering:** Remove or enhance certain frequency components of the signal



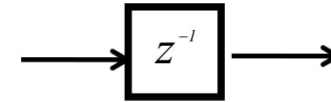


Key components of a block diagram

1. The Ideal Delay

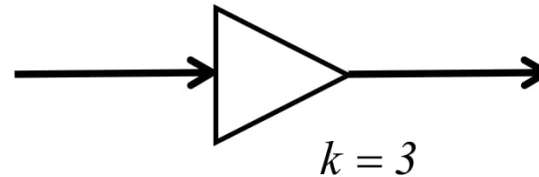
$$y[n] = x[n - 1], \quad (2.6)$$

The sample will shift to the right
for one time instant



Unit delay

2. Amplifier/Gain/Multiply by constant



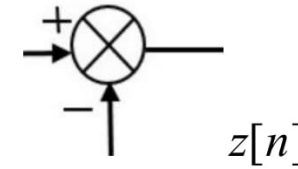
The output of a gain only depends on current input. From the diagram, the output will be $k = 3$ times the magnitude of the input.



3. Summation

$$y[n] = x[n] - z[n] \quad (2.7)$$

- Pay attention to the + - sign at the summation point.



4. Downsampler

$$y[n] = x[Mn], \quad -\infty < n < \infty \quad (2.8)$$

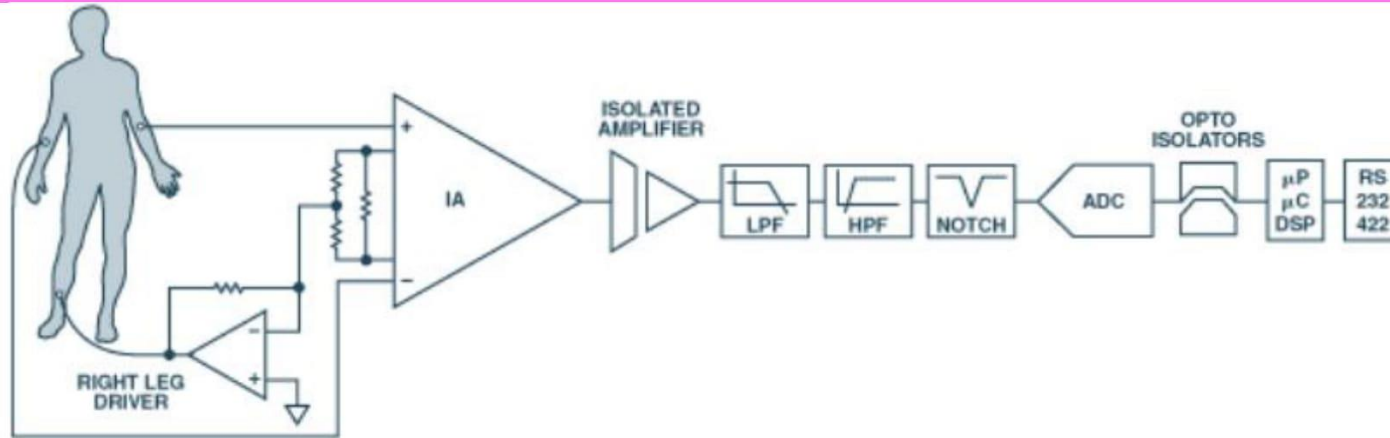
M is a positive integer. It discards $(M - 1)$ samples every M samples. It is not time-invariant unless $M = 1$.



5. Upsampler

Performs the opposite of downsampler.

Example 2.1 ADC in ECG Holter system Frontend (Cont.)



- The whole circuit can be regarded as a discrete-time system.
- The mathematical form of $T\{x[n]\}$ could be quite complicated.
- It involves a cascade of amplification, low pass, high pass, notch filter, analog-to-digital conversion, before reaching the digital signal processor (DSP) and the serial port (RS232, 432).



1.5 Linear systems

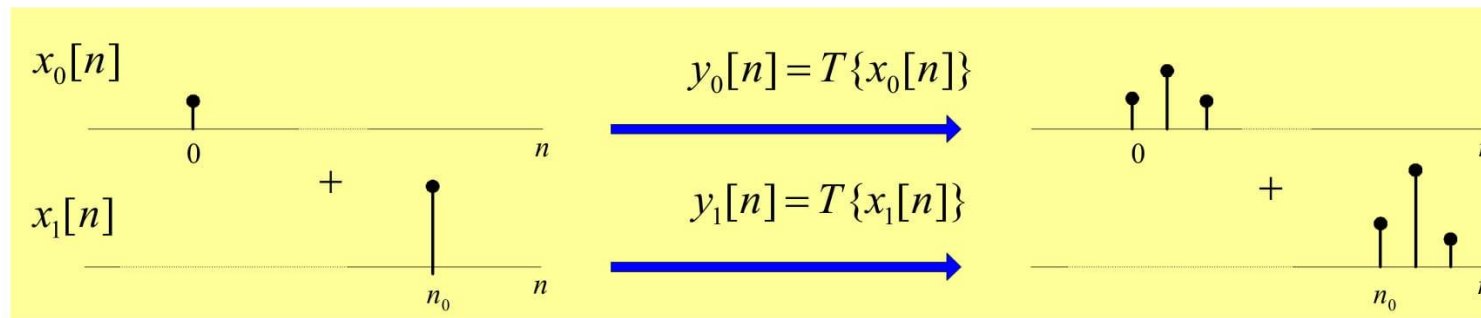
Linear systems satisfy the principle of superposition:

Let $y_1[n]$ and $y_2[n]$ be the responses of a system when the inputs are respectively $x_1[n]$ and $x_2[n]$. The system is linear if and only if

$$T\{x_1[n] + x_2[n]\} = T\{x_1[n]\} + T\{x_2[n]\} \quad (2.9) \quad \text{(additive property)}$$

$$\text{and } T\{ax[n]\} = aT\{x[n]\} = ay[n] \quad \text{(scaling property)}$$

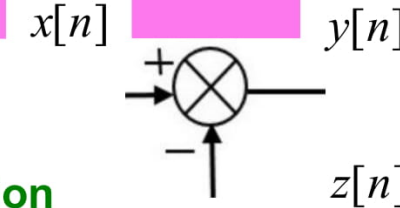
where a is an arbitrary constant.





Example 2.2 Summation

$$y[n] = x[n] - z[n].$$

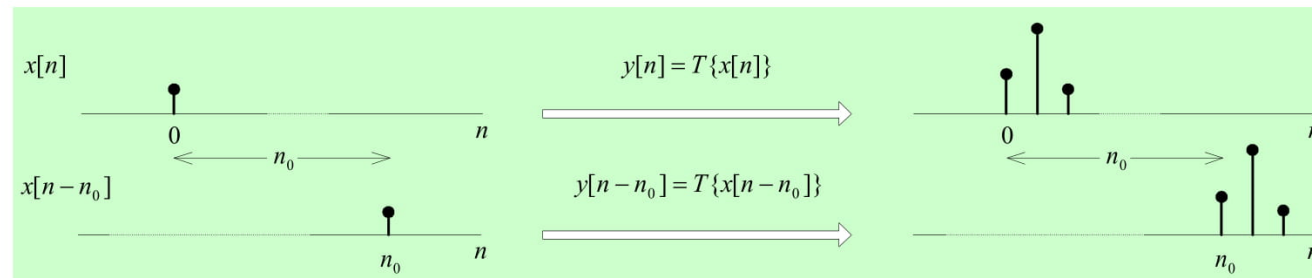


- Pay attention to the + - sign at the summation point.

1.6 Time-invariant system

A **time-invariant system** is one for which a time shift of the input sequence causes a corresponding shift in the output sequence, i.e.

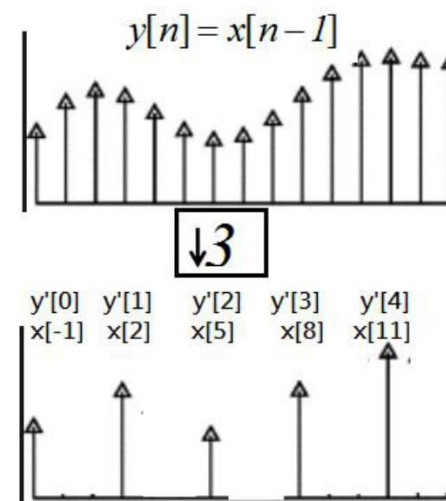
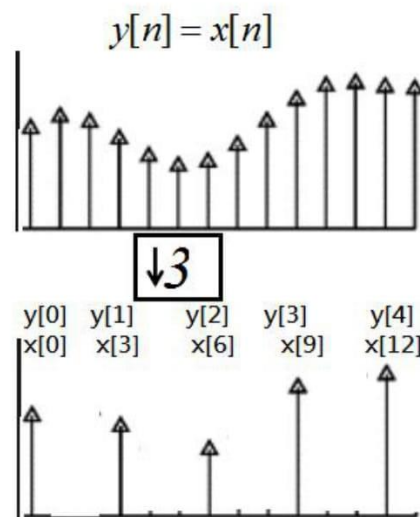
If $y[n] = T\{x[n]\}$, **then** $y[n - n_0] = T\{x[n - n_0]\}$ **for all integer** n_0 . (2.10)



**Example 2.3 Downsampler (Not time-invariant)**

$$y[n] = x[Mn], \quad -\infty < n < \infty \quad (2.11)$$

M is a positive integer. It discards $(M - 1)$ samples every M samples. It is not time-invariant unless $M = 1$.



$$y[n-1] \neq y'[n] \quad \text{!!!! e.g. } y[2] = x[6] \neq y'[3] = x[8]$$



1.7 Causality

- For causal system, $h[n] = 0$ for $n < 0$.
- Relies on current and previous samples only.

Example 2.4 Forward Difference System

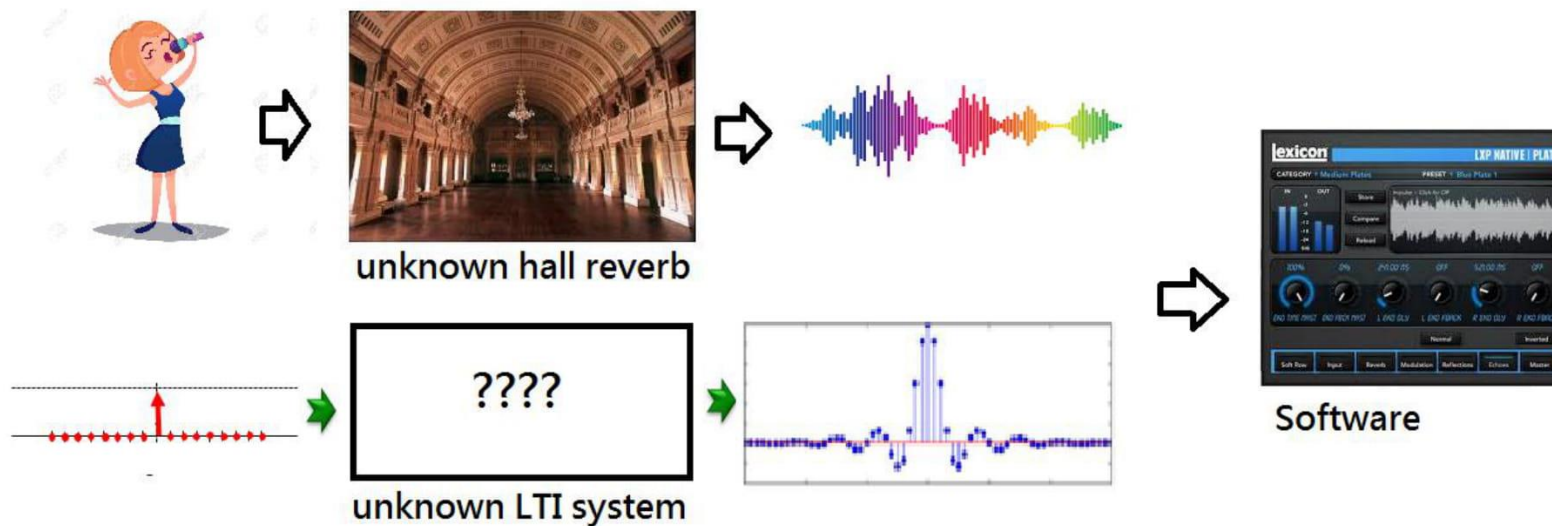
$$y[n] = x[n+1] - x[n]$$

(2.12)

The system is non-causal (finitely non-causal) because it involves a future input sample $x[n+1]$.



2. FIR and IIR SYSTEMS



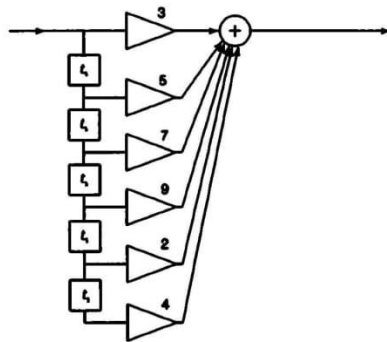
- Suppose we want to create a digital filter to reproduce the hall reverb.
- FIR filters or IIR filters can be used for such purpose.

**Finite-duration impulse response (FIR)**

$$y[n] = \sum_{k=0}^M h[k] \cdot x[n-k] \quad (2.13)$$

M is called the **order/degree** of the system.

$M+1$ is the **filter length**. No feedback !



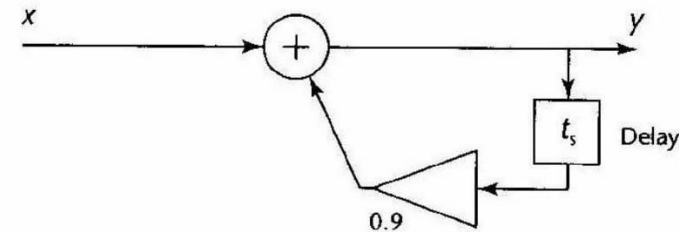
$$y[n] = 3x + 6x[n-1] + 7x[n-2] + 9x[n-3] + 2x[n-4] + 4x[n-5]$$

Infinite-duration impulse response (IIR) filter

Example:

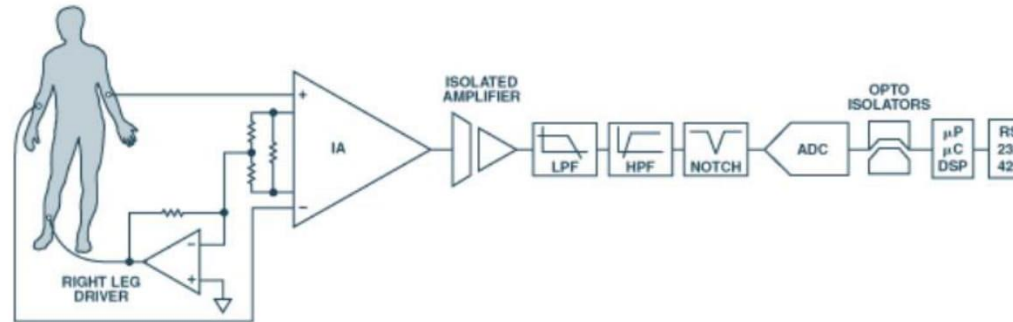
$$y[n] = a \cdot y[n-1] + x[n]. \quad (2.14)$$

There is feedback !



Applications to Biomedical Signal Processing

Example 2.1 ECG Holter system Frontend (cont)

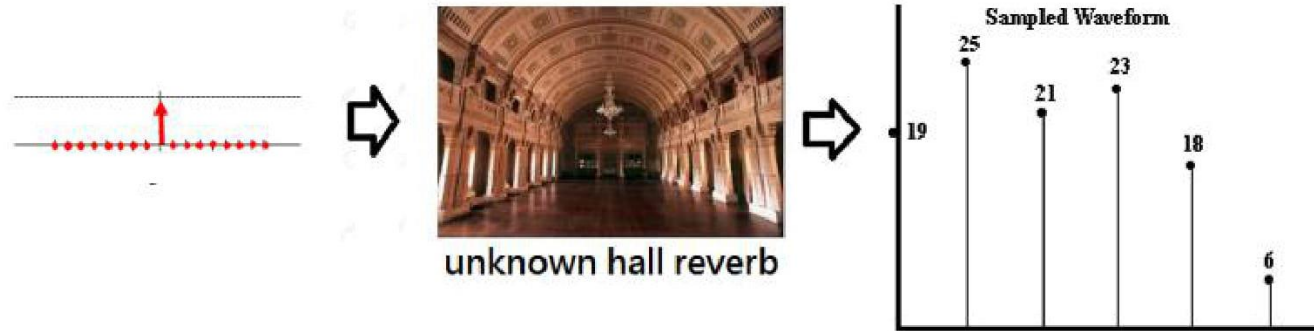


The Low-pass filter (LPF), high-pass filter (HPF) and notch filter can be designed either by FIR or IIR filters.



2.1 Filter Design (Windowing approach)

- Suppose we want to create a digital filter to reproduce the hall reverb.

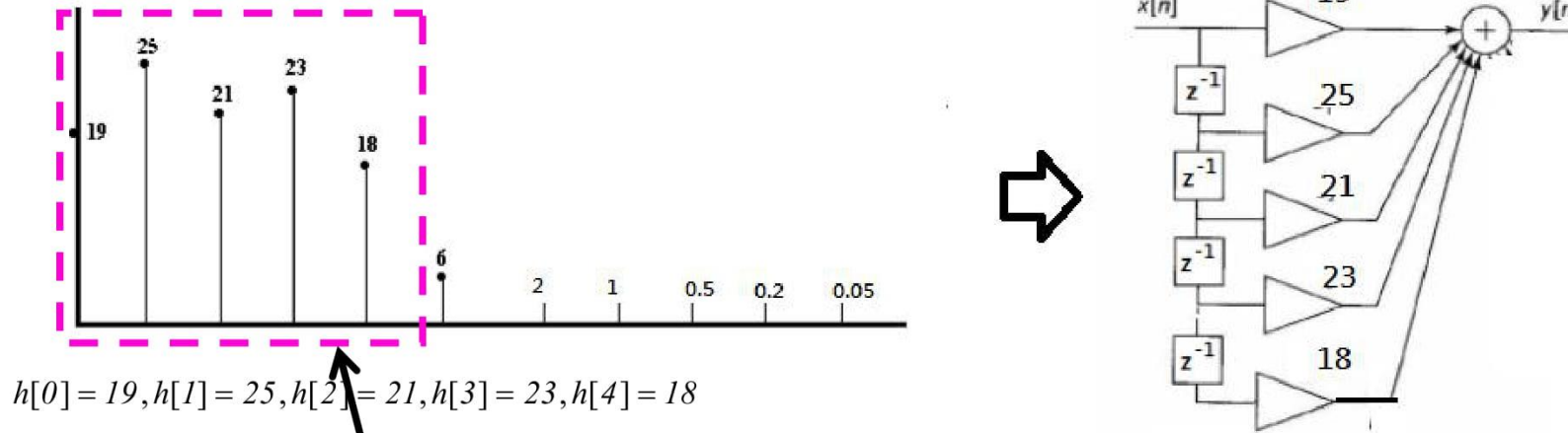


1. First, feed an impulse (continuous-time) through the hall environment.

E.g. clap your hand 🖐️

2. Sample the waveform obtained from the hall using a USB ADC microphone.

Fig. 2.5 Windowing approach



- This obtains the impulse response $h[n]$ of the hall reverb.
- We only keep the beginning part and **truncate** the rest when $h[n]$ is long.
- This is called **windowing**.
- The gains of the FIR filter are also called **filter coefficients**.
- **Requirement:** The system has to be **linear time-invariant**.



2.2 Linear time-invariant systems and impulse response

What is impulse response?

It is the output of a system when presented with an impulse/unit sample sequence $\delta[n]$

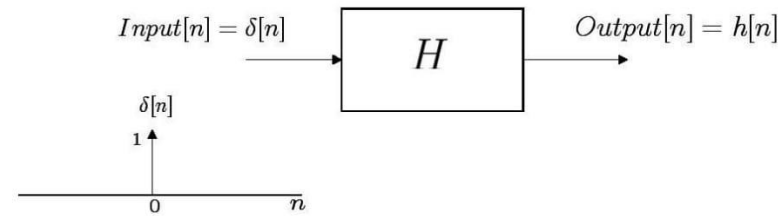


Image source http://signalsandsystems.wdfiles.com/local--resized-images/impulse-response/ch2_1_1.jpg/medium.jpg

A linear system is completely characterized by its impulse response

Let $h_k[n] = T\{\delta[n - k]\}$ be the response of the system to $\delta(n - k)$, an impulse occurring at $n = k$. A linear time-invariant system can be given as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] = x[n] * h[n] \quad (2.15)$$

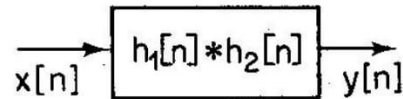
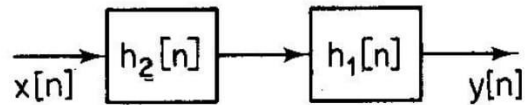
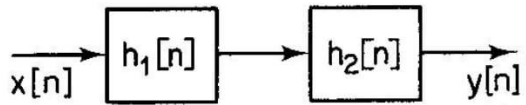
* denotes discrete-time convolution.



2.2 Properties of LTI systems

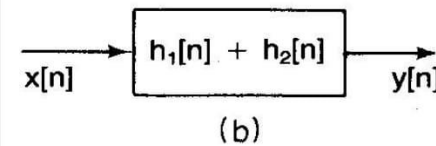
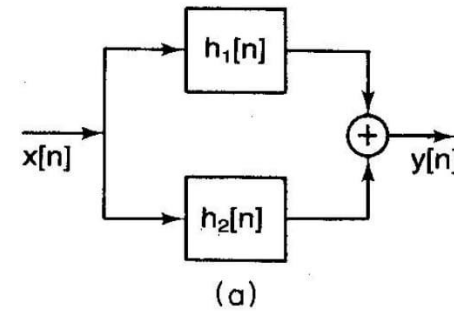
1) Systems in Cascade:

$$h[n] = h_1[n] * h_2[n] \quad (2.20)$$



2) Systems in Parallel

$$h[n] = h_1[n] + h_2[n] \quad (2.21)$$





2.3 Difference equations

We use a N^{th} -order linear constant-coefficient difference equation to describe a discrete-time system with input $x[n]$ and output $y[n]$

$$y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k] \quad (2.19)$$

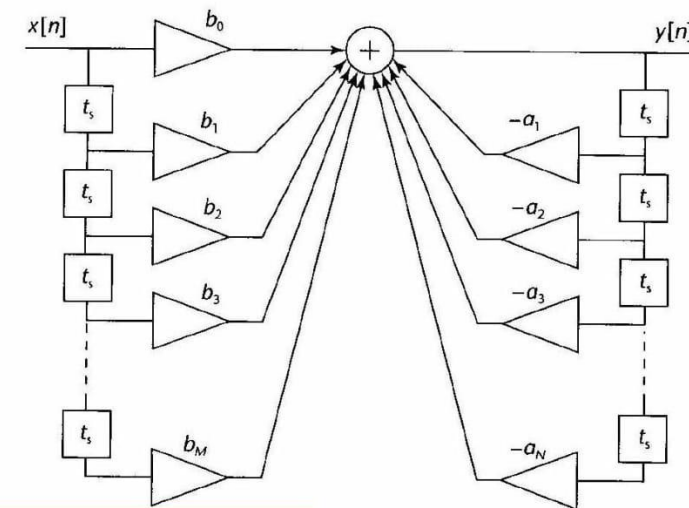
Feedback

Feedforward

Continuous-time (ODE) (Example)

$$y(t) = -\frac{d^2 y}{dt^2} - 10 \frac{dy}{dt} + 4 \frac{dx}{dt} + x(t)$$

Compare between the ODE and difference equation



Feedforward (non-recursive) part

Feedback (recursive) part

Structure of a recursive (IIR) digital network

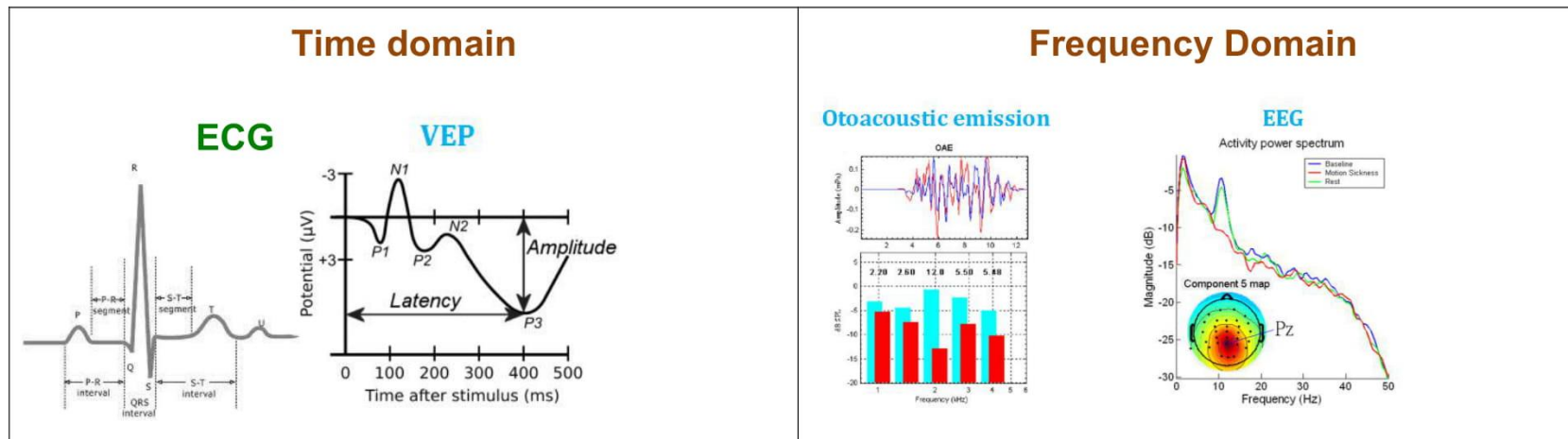


4. FREQUENCY RESPONSE AND VARIOUS TRANSFORMS

4.1 Signal Representation

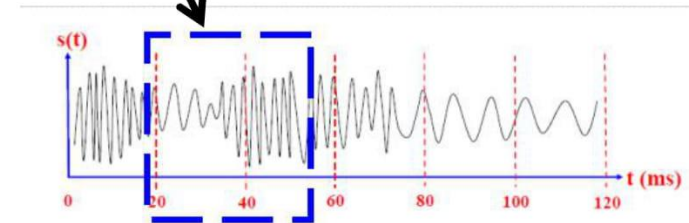
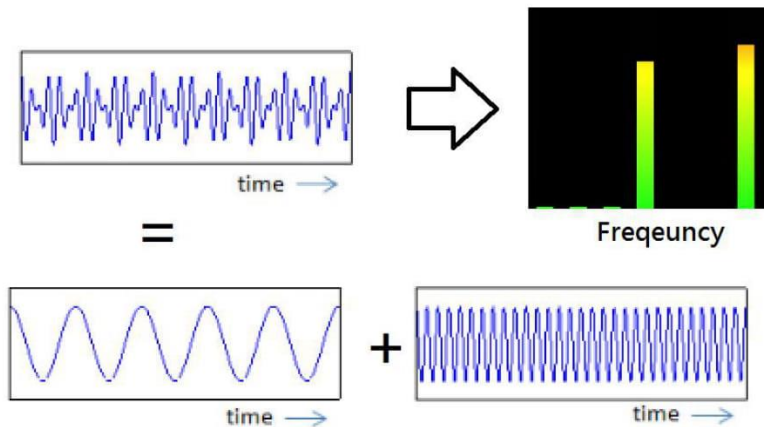
■ Different biomedical signals may be better presented in the following domains:

1. time domain or temporal domain
2. frequency domain or spectral domain
3. space domain or spatial domain



4.2 Fourier Transformation

- A stationary waveform can be decomposed into different sinusoidal components.
- By converting them to frequency domain,
- we are able to visualize the different components of the signal more easily.
- For non-stationary waveforms, we can capture windows of them and convert them to frequency domain.





- There are 3 extra transforms that we learn in this chapter.

Continuous-time fourier transform (CTFT)	$F(j\Omega) = \int_0^\infty f(t)e^{-j\Omega t} dt$ is a function (continuous waveform)
Discrete-time fourier transform (DTFT)	$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$ is a function (continuous waveform)
Discrete fourier transform (DFT)	$X[k] = X(e^{j\omega_k}) = \sum_{n=0}^{N-1} x[n]e^{-j(2\pi nk / N)}$ is a sequence (discrete)

- **CTFT** converts a continuous-time signal into frequency domain. The output is a continuous waveform. The frequency ranges from $\Omega \in [-\infty, +\infty]$
- **DTFT** is a discrete-version of CTFT. CTFT is an integral. DTFT reduces to an infinite sum. $\omega \equiv (\Omega / f_s) \bmod 2\pi$ is the **digital radian frequency**.
- **DFT** are uniform samples of the DT-FT in the frequency domain (spacing $2\pi / N$).



Vector Space

Basis:

Basis vector are orthogonal

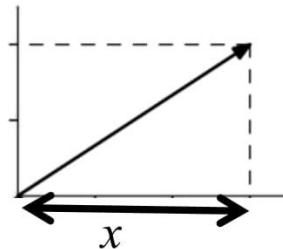
E.g. $i \cdot j = 0$

Calculation of basis coefficient:

E.g. Finding the i component for $[1, 3, 5]^T$

$$[1, 3, 5]^T = xi + yj + 5k$$

$$x = [1, 3, 5]^T \cdot i = 1$$



Functional space

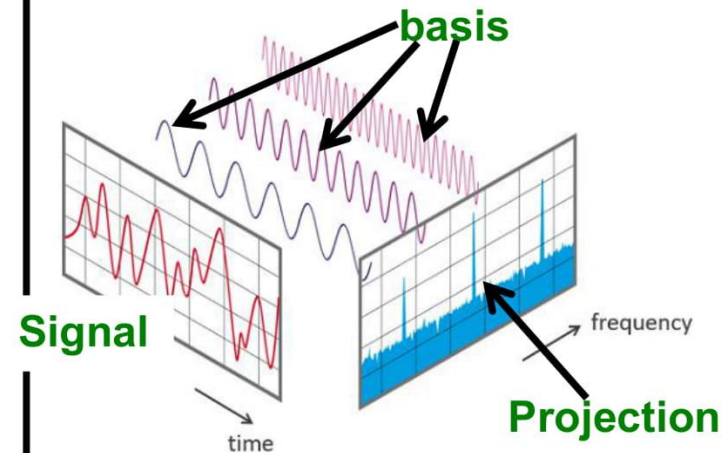
Basis:

$$\int_0^T \Phi_1(t) \Phi_2^*(t) dt = 0 \text{ (orthogonal functions)}$$

E.g. $\int_0^T e^{j50t} (e^{j40t})^* dt = \int_0^T e^{j\frac{2\pi mt}{T}} (e^{j\frac{2\pi nt}{T}})^* dt$

(Proof left as exercise)

Calculation of basis coefficient



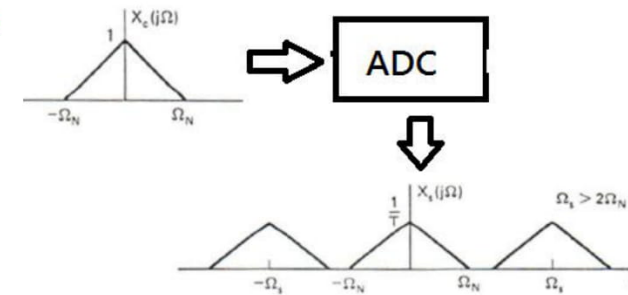


5. SAMPLING THEOREM

5.1 Digital Radian Frequency

- During analogue to digital conversion, the signal is sampled at frequency $f_s = \frac{1}{T_s}$.

$$x[n] = x_c(nT_s), -\infty < n < \infty$$



- Suppose the frequency of the original signal ranges from $\Omega \in [-\infty, +\infty]$,
- $\omega \equiv \frac{\Omega}{f_s} \bmod 2\pi$ is called **digital radian frequency**, where f_s is sampling frequency. The range of ω is $[-\pi, \pi]$.
- The mod operator returns the remainder after dividing by 2π .



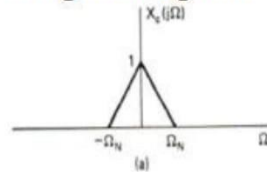
- The 'mod' operator indicates the copies of $X(j\Omega)$ will repeat at integer multiples of the sampling frequency.
- If the input is bandlimited to Ω_N (maximum frequency of signal) and

$$|\Omega_s| > 2\Omega_N$$

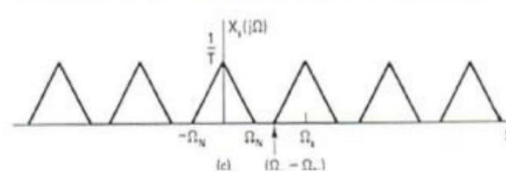
(sampling theorem is satisfied),

Then there will be no aliasing. Otherwise, there will be aliasing.

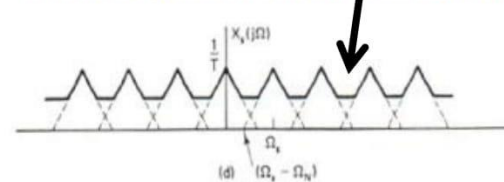
Original Signal



Sampling frequency $> 2 \times$ max. freq.

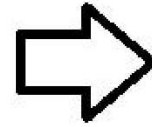
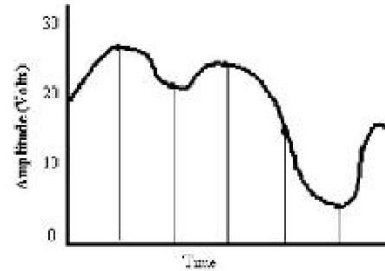


Sampling frequency too low (aliasing)

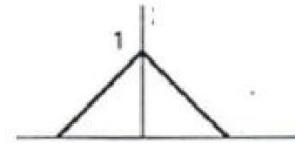




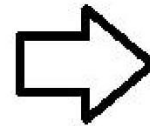
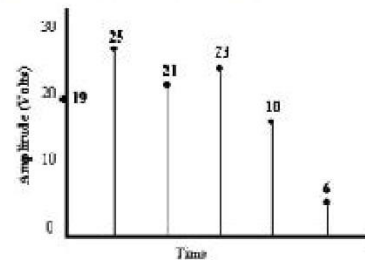
Continuous waveform



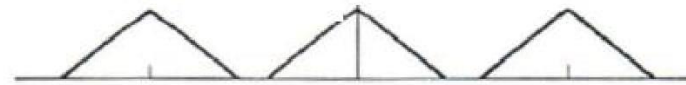
CTFT (Continuous function)



Discrete waveform



DTFT (Continuous function)



DFT

Sampled by N-point

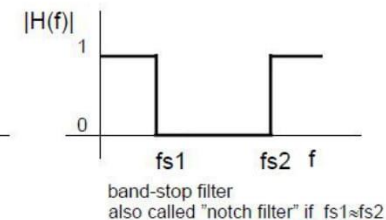
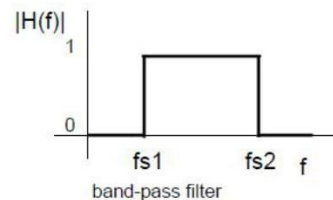
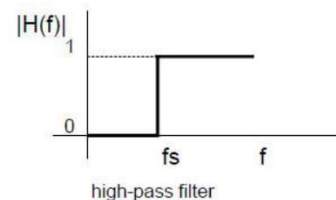
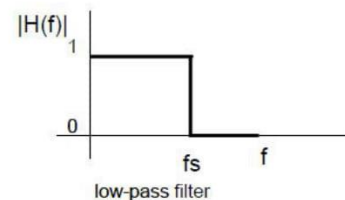




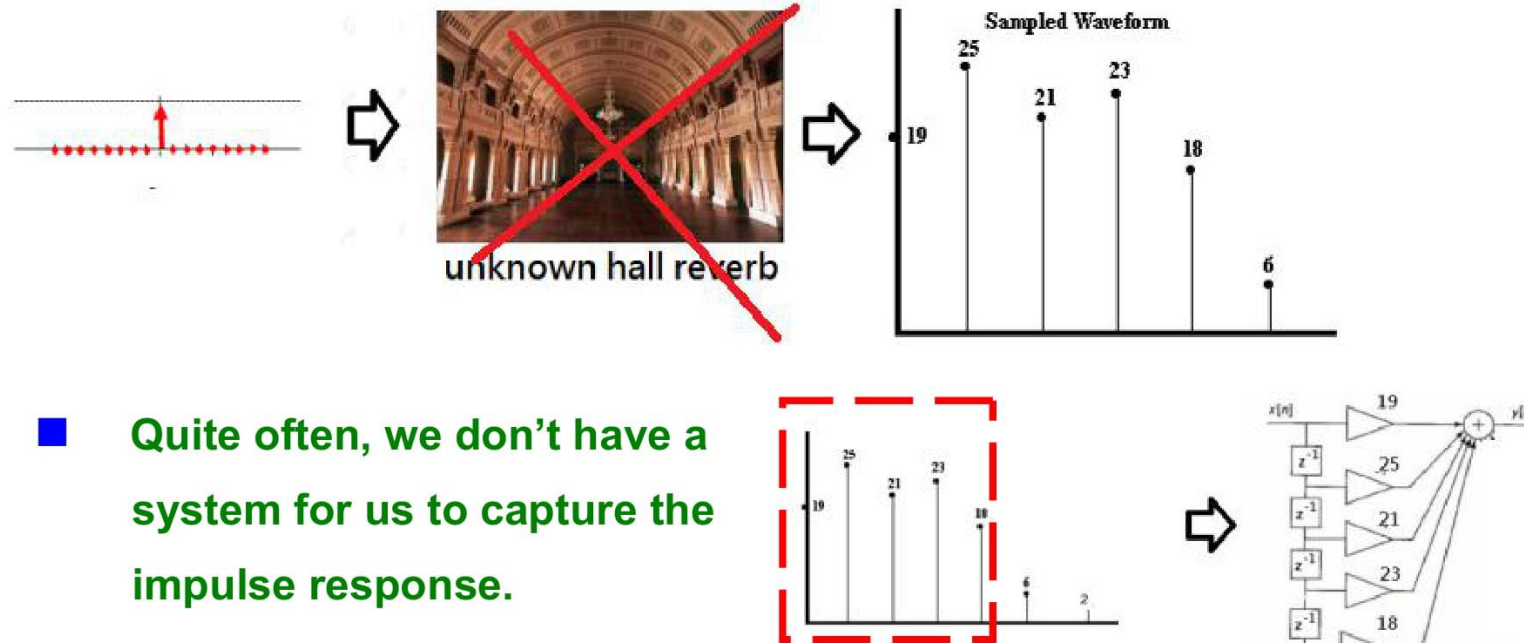
6. FILTER DESIGN

6.1 Types of Filter

- **Lowpass filter:** to remove high-frequency noise (for example white noise, EMG in EEG)
- **Highpass filter:** to remove low-frequency noise (for example, trends, drifts, artifacts)
- **Bandpass filter:** to remove both low and high frequency noise; can be a cascade of lowpass and highpass filter
- **Bandstop filter:** to remove narrow-band noise (for example notch filter to remove powerline interference); can be a cascade of lowpass and highpass filter



6.2 FIR Filter Design according to Specification



- Quite often, we don't have a system for us to capture the impulse response.

- We have to begin with certain specification on the frequency range.



FIR Vs IIR Filters

FIR Filters	IIR Filters
Contains only zeros (always stable)	Contain zeros and poles
Can have exact linear phase response -> No phase distortion	Phase response is usually non-linear, especially at band edges
Quantization effects are less severe	After coefficient quantization, a pole can move to outside of the unit circle (cause instability)



Sharp cutoff and low passband delay requires long filter length.	Potentially lower filter order for low passband delay and sharp cutoff.
Optimal design using Parks-McClellan algorithm or SOCP (linear and passband linear phase).	Require nonlinear optimization. Perform model order reduction (MOR) of a long optimal FIR filter.

- **Linear phase** (the same time delay for all frequencies) is often **important** for biomedical signal interpretation, as it keeps the time relationships between different waves intact.
- A smaller order is preferred for a shorter delay in online processing.

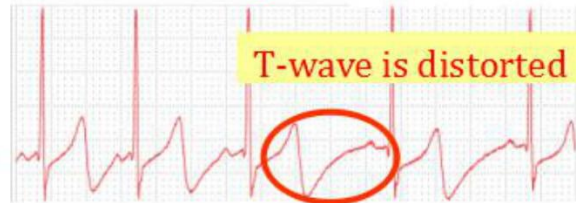


Example 2.8: Filtering of ECG

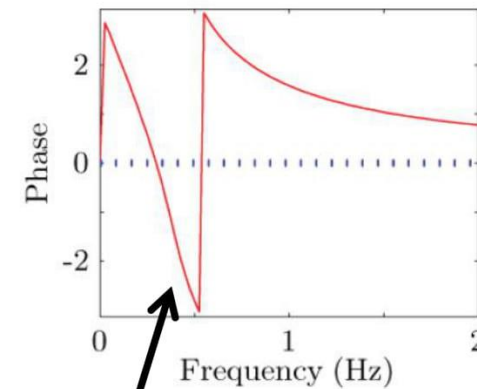
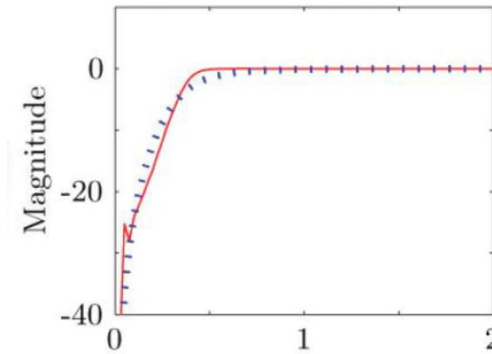
Desired Signal



IIR Filtered Signal



FIR Filtered Signal



IIR: Non-linear Phase

FIR: Linear Phase